Retail Inventory & Demand Forecasting Analysis

Project Performance Summary

Executive Key Performance Indicators (KPIs)

10M	$550M	$55M	10M	$55
Total Units Sold	Total Sales Revenue	Discount Amount	Demand Forecast	Avg Price

The project maintains a healthy equilibrium between actual units sold and demand forecast, both standing at approximately 10 million units. The revenue-to-discount ratio is approximately 10:1, indicating a controlled promotional strategy.

Revenue Analysis

1. Regional Distribution

Sales revenue is remarkably balanced across four major regions, with each contributing approximately 25% of total revenue:

- **West:** $138.71M (25.21%)
- **East:** $138.48M (25.17%)
- **South:** $137.26M (24.95%)
- **North:** $135.78M (24.68%)

2. Seasonality Impacts

Sales show a consistent upward trend through the seasonal cycle, with **Summer** contributing the final push to reach the $0.55bn total. Each season contributes roughly $0.14bn to the total revenue figure.

Product Category Insights

The Category Sales Volume tree map reveals a fairly even distribution across major categories, with **Furniture**, **Groceries**, **Clothing**, and **Electronics** all generating revenue in the range of $109M to $111M.

Actual vs Forecasted Sales by Category

Forecasting accuracy is high across all segments. Furniture leads in volume (approx. 2.03M units), followed closely by Groceries and Clothing. Even the lowest performing category (Electronics) maintains a high volume at 1.96M units.

Performance Matrix Preview

Category	Seasonality	Sum of Discount	Sum of Units Sold
Clothing	Autumn	36,385	498,364
Clothing	Spring	36,830	500,879
Electronics	Autumn	36,460	492,630
Electronics	Summer	36,475	490,770

*Detailed breakdown indicates consistent unit sales across seasons per category, suggesting stable year-round demand.

Conclusion

The analysis demonstrates a highly stable retail environment with minimal regional variance and consistent demand. The alignment between actual sales (10M) and forecast (10M) suggests an extremely accurate predictive model is currently in place.